

Non-linear Semantic Decoupling for Few-shot Object Detection

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Abstract

Once few-shot annotated samples are available, the performance of learning-based object detection would be heavily dropped. Many few-shot object detection (FSOD) methods have been proposed to tackle this issue by adopting image-level augmentations in linear manners. Nevertheless, those handcrafted enhancements often suffer from limited diversity and lack of semantic awareness, resulting in unsatisfactory performance. To this end, we propose a non-linear instance-level data augmentation method by decoupling the foreground and background to increase their diversities respectively. We design a semantic awareness enhancement strategy to separate objects from backgrounds. Concretely, masks of instances are extracted by an unsupervised semantic segmentation module. And then the diversity of samples would be improved by fusing instances into different backgrounds. Considering the shortcoming of augmenting images in a limited transformation space of existing traditional data augmentation methods, an object reconstruction enhancement module is proposed to generate sufficient diverse and non-linear training data at the instance level by the self-supervised masked autoencoder. In this way, the potential of data can be fully exploited in various object detection scenarios. Extensive experiments demonstrate two different models equipped with our idea significantly outperform baselines by a large margin. New state-of-the-art results on benchmarks of the PASCAL VOC and MS COCO under different shot settings are also achieved. Source codes are included in supplementary files.

1. Introduction

Over the past few years, deep learning based methods have achieved state-of-the-art performance in a variety of visual tasks, such as object recognition [36, 39], and image segmentation [15, 30]. However, it relies heavily on abundant annotated data, which limits their applicability to some realistic scenes, where samples are hard to collect or annotations are expensive. In contrast, humans can quickly grasp a novel concept with few labeled samples. To bridge

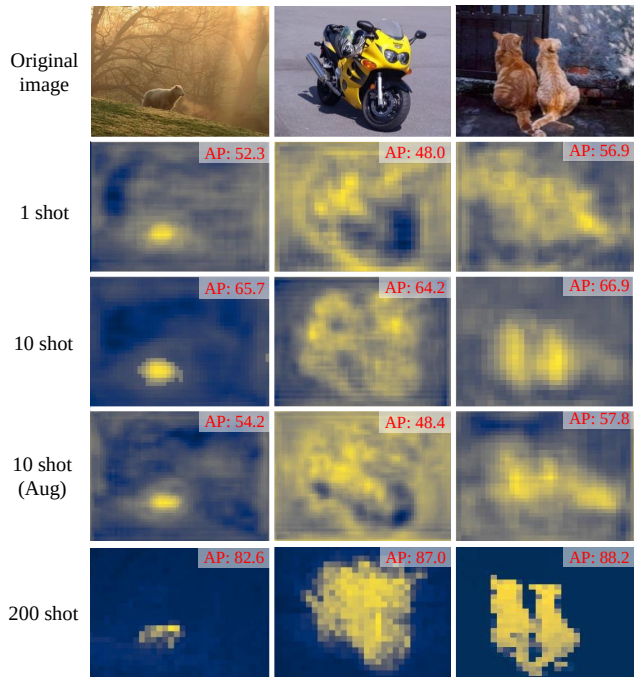


Figure 1. Illustration of confidence weighting maps under different shot settings on PASCAL VOC. N-shot means each category with N annotations during finetuning. 10 shot (Aug) indicates data augmentation from 1-shot to 10-shot with traditional linear augmentation methods. Once more annotations are available, foreground features would be salient and better separated from the background (row.2 to row.5). Although traditional data augmentation methods scale up the sample amount, they are inadequate in effectively enhancing the sample diversity (row.3 vs row.4), resulting in unsatisfactory performance improvement.

the gap between the performance of deep learning and the ability of humans, few-shot learning [25, 33, 35, 40, 44] has attracted much attention.

Though considerable progress has been recently made, most previous works [4, 27, 28] mainly focus on the few-shot classification task. However, relatively little effort has been made in the more challenging task, *few-shot object detection* (FSOD) [8, 34, 46], which aims to predict not only

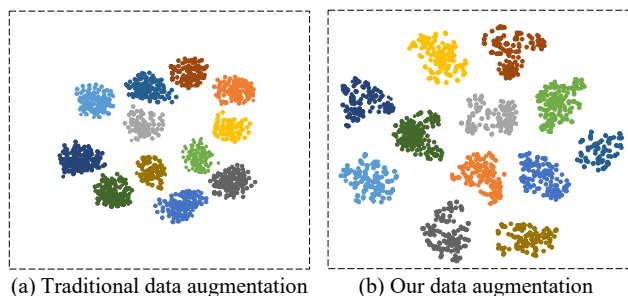


Figure 2. The t-SNE [42] visualization of object proposal embedding learned with (a) traditional data augmentation method and (b) our proposed method respectively. It shows the results of augmenting randomly selected 200 PASCAL VOC images to 2400 images. Instances from each category form tight clusters, indicating their similarities. In comparison, clusters generated by our method are much more divergent than ones by the traditional linear method. That is, samples augmented by our method would have much greater diversities, resulting in a better FSOD performance.

class predictions but also box regression from only a few training samples. FSOD, typically with only K instances per class, easily lacks generalization capability and incurs overfitting since only extremely few annotated instances of previously unseen categories are available for training.

An obvious solution for FSOD is data augmentation [22, 23], which tries to increase the diversity of the input information by directly expanding the number of training labeled instances or indirectly augmenting the features. Most existing data augmentation methods [3, 5, 20, 21, 24, 41, 49] obtain more samples through handcrafted enhancements (e.g. flipping, rotating, scaling, cutting) at levels of images, proposals, or features. However, the common characteristic of these methods is that linear-based augmentations lack semantic awareness and can not generate various training samples with sufficient diversity, since they are not foreground/background sensitive.

As shown in Figure 1, for a ResNet-101 [16] based object detection model, the confidence weighting maps would gradually become salient once more training instances for novel classes are available. That is, when there are only a few instances, information distinguishing foreground and background tends to be noisy and local (Figure 1, row.2 and row.3). Compared with few-shot instances, adequate annotations could help converge to salient features and achieve better separation from the background (Figure 1, row.2 vs row.5). At present, there is no practical solution to obtain semantic information for few-shot instances. For example, though augmenting 1-shot to 10-shot with traditional methods, such as flipping/rotating/scaling, increases the sample quantity, the performance improvement is limited since the semantic information is not well enhanced and the foreground/background still can not be readily separated (Fig-

ure 1, row.3 vs row.4), resulting more susceptible to overfitting due to the lack of diversity.

In this paper, a novel data augmentation method is proposed to address these issues for FSOD, which decouples the foreground and background to increase their diversity and augments various training data in non-linear transformation spaces. It consists of the semantic awareness enhancement strategy (SAES) and the object reconstruction enhancement module (OREM) to generate augmented training data at the instance level. The SAES is adopted to decouple foreground and background. And then the diversity of samples would be improved by fusing instances into different backgrounds, so as to further enhance the semantic awareness of the object’s distinct features. To avoid expanding images in a limited linear transformation space, inspired by the powerful nonlinear fitting and reconstruction capabilities of the masked image modeling [14], we introduce OREM to generate diverse data for few-shot samples in a non-linear manner. It randomly masks instances of the novel classes and feeds them into a self-supervised encoder-decoder architecture to export reconstructed images.

As illustrated in Figure 2, the t-SNE [42] visualization shows the proposal embedding of augmenting randomly selected 200 PASCAL VOC images to 2400 images by traditional and proposed methods respectively. Instances from each category form tight clusters, indicating their similarities. In comparison, clusters generated by our method are much more divergent than ones by the traditional linear method. That is, samples augmented by our method would have much greater diversities, resulting in a better FSOD performance. To summarize, our contributions are 3-fold:

- (1) In this paper, we present a novel data augmentation method for FSOD, which decouples the foreground and background of the novel class objects and increases their diversity in a non-linear manner respectively.
- (2) The core idea of our method is non-linear semantic decoupling. Thus, the SAES is adopted to decouple foreground/background and the OREM is introduced to generate diverse data for novel classes in a non-linear manner.
- (3) The proposed method can be readily plugged into existing finetuning-based FSOD methods for further performance improvement. Experiments on two different models equipped with our idea significantly outperform baselines by a large margin. New state-of-the-art results on FSOD benchmarks of the PASCAL VOC and MS COCO under different shot settings are also achieved, demonstrating the effectiveness and robustness of our idea.

2. Related Work

2.1. Few-Shot Object Detection

Differing from general object detection [9, 10, 36, 38, 39] with adequate annotations, few-shot object detection (FSOD) [2, 7, 19, 47, 53] aims to detect objects with few

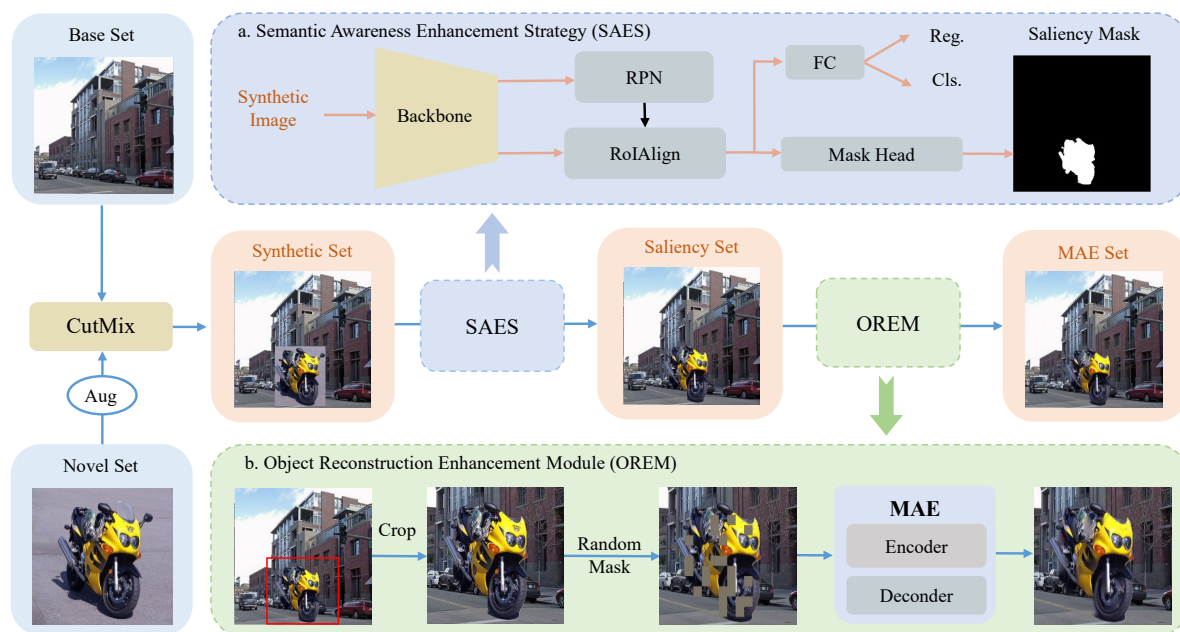


Figure 3. The overall framework. To enhance the generalization ability, we present a novel data augmentation method consisting of two steps to generate diverse data. (a) To decouple the foreground/background of few-shot instances, Semantic Awareness Enhancement Strategy (SAES) is adopted to extract the mask of instances. And then the diversity of samples would be improved by fusing instances into different backgrounds. (b) Object Reconstruction Enhancement Module (OREM) is employed to randomly mask instances and feed them into MAE [14] to export the enhanced image in a non-linear manner.

labeled samples, which could be roughly categorized into the meta-learning fashion and finetuning fashion. For the meta-learning fashion, FSRW [18] leverages a reweighting module with adequate labeled base classes based on YOLOv2 [37] and transfers to novel classes quickly. FS-DetView [50] introduces a joint feature embedding to make the most of rich feature information originating from base class sharing. These methods usually suffer from the design of complex episodic training schemes. Recently, several finetuning methods [34, 46] attracted more attention compared to meta-learning methods. Two-stage Finetuning Approach [46] (TFA) is the baseline method of finetuning fashion, which trains on fully labeled base classes in the first stage, and finetunes on the balanced support set in the second stage. A recent work, DeFRCN [34], decouples conflict tasks between class-agnostic RPN and class-relevant RCNN. However, the overfitting problem easily arises in this fashion with limited instances of novel classes. In this work, we design a novel non-linear instance-level data augmentation method to alleviate the problem.

2.2. Data augmentation in FSOD

Data augmentation is a common tool for improving performance in computer vision, especially in cases where training data is not abundant, such as FSOD. In the recent literature, MPSR [49] and FSOD-SR [21] generate multi-

scale positive instances as object pyramids to solve the problem of scale variations. LVC [20] introduces a pseudo-labeling method to source high-quality pseudo-annotations in novel categories. FSCE [41] regards the proposals of different IoU as the intra-image augmentation used in contrastive methods. However, the above linear augmentation methods enhance sample quantity in the limited transformation space, and cannot generate sufficient diversity of data.

NP-RepMet [52] incorporates the negative proposals discarded into the model training, which results in a more robust embedding space. To address the lack of variability in the training data, Halluc [57] introduces a hallucinator network to transfer the shared within-class variation from base classes to novel classes. And TIP [24] introduces transformed guidance consistency loss on predictions from various transformed images. However, most of the above methods rely on image-level augmentation and lack of semantic awareness, resulting in unsatisfactory performance. Different from previous methods for FSOD, our method improves the diversity of samples by non-linear semantic decoupling at the instance level.

3. Method

3.1. Problem Setting

We first introduce the problem setting of few-shot object detection. Referring to the problem setup [18, 46, 50]

used in previous research, we split the general object detection datasets into the few-shot object detection datasets. Specifically, the object class is divided into the base class set C_{base} with abundant annotations and the novel class set C_{novel} with only k labeled samples in each class, where C_{base} from the base class dataset d_{base} , C_{novel} from the novel class dataset d_{novel} . There is no intersection between two category sets, $C_{base} \cap C_{novel} = \emptyset$. Our approach follows the finetuning-based methods of FSOD, which can be divided into the base training stage and the novel class finetuning stage. In the first stage, the base class with adequate annotations is used to train the model, so that the detector has the information that can be transferred to the novel class. In the second stage, finetuning the novel class with scarce samples to make the whole detector more general.

3.2. Overview

The overall proposed framework is shown in Figure 3. We present a non-linear instance-level data augmentation method for FSOD. Our approach first applies CutMix data augmentation [54] to produce a synthetic set (d_{syn}), which crops entire object patches (P_n) from C_{novel} images and randomly pastes them onto C_{base} images. Next, we adopt the Semantic Awareness Enhancement Strategy to generate a saliency set ($d_{saliency}$), which obtains semantic awareness through foreground/background decoupling and fusing instances into different backgrounds. To further increase the diversity of samples, we employ the Object Reconstruction Enhancement Module to generate the MAE set (d_{mae}) during the finetuning stage, which uses self-supervised masked image modeling to generate augmented images of few-shot instances. The proportion of each generated set during finetuning is determined based on a predefined ratio as follows:

$$d_{aug} = k_1 d_{syn} + k_2 d_{saliency} + k_3 d_{mae} \quad (1)$$

Where the hyperparameter k_1 , k_2 , and k_3 represent the proportion of different generated sets, thereby regulating the tradeoff between them.

3.3. Semantic Awareness Enhancement Strategy

To increase the number of novel class samples, we perform the CutMix data augmentation to combine base class images with novel class objects, which crops entire object patches (P_n) from images in d_{novel} and applies simple random data augmentation techniques such as scaling, flipping, and color degradation. The augmented object patches are then pasted onto the images I_b from d_{base} . By repeating this process, we generate a synthetic set (d_{syn}) containing samples from both d_{novel} and d_{base} . The visualization of d_{syn} is shown in Figure 4 (col.1).

However, foreground and background coupling in image-level augmentation methods such as CutMix makes

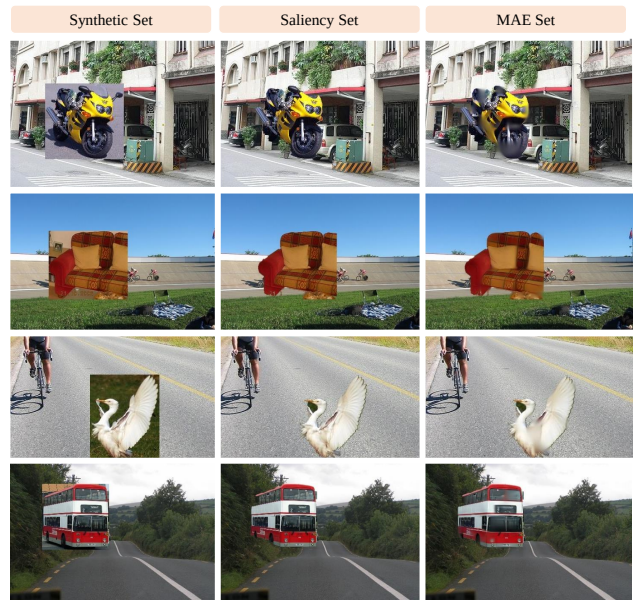


Figure 4. Visualizations of different sets generated using the base set and novel set of PASCAL VOC. The synthetic set and saliency set are generated by SAES, which augment the number of samples. The diversity would be improved by fusing instances into different backgrounds. MAE set constructed by OREM further implements instance-level non-linear enhancement. Best viewed with zoom in.

the boundaries not well augmented, resulting in limited diversity of objects. To address this issue, we adopt the Semantic Awareness Enhancement Strategy (SAES), to guide the network by decoupling the foreground and background to extract object semantic discriminative features during finetuning. In our approach, we add a split branch (Mask Head) to our baselines to learn the mask of C_{novel} samples. The input to the network is the synthetic set d_{syn} generated by CutMix, and the output is the detection results of all objects as well as the mask of novel class samples.

As there are no semantic masks available for few-shot objects, we need to rely on unsupervised semantic segmentation methods. However, learning dense semantic representations of few-shot objects in an unsupervised setting is a challenging task, as the network may prioritize low-level image features over image semantics in an extremely low-data regime. MaskContrast [43] proposes an unsupervised semantic segmentation method, which obtains the mask from unsupervised saliency detection as a prior. Its key idea is to contrast the features of the pixels within the object mask proposals with those in the background and to use this contrast as a criterion for segmenting the objects. Motivated by the above method, the unsupervised saliency detection [32] is also introduced to learn the mask regions of novel class objects and apply it as a prior for guiding the network to learn the representation of pixels for object

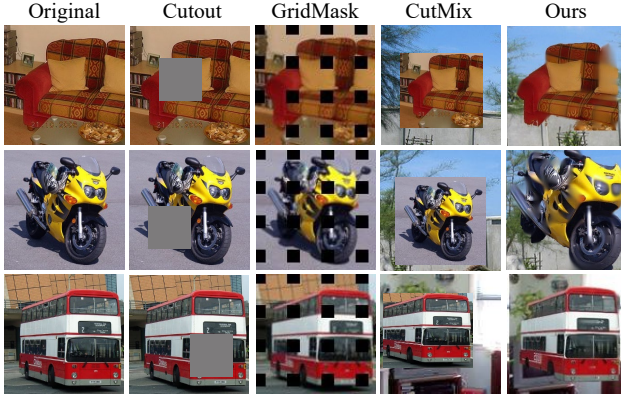


Figure 5. Visualizations of the results of Cutout [5], GridMask [3], CutMix [54], and our data augmentation method on Pascal VOC. Unlike the above methods, our method obtains semantic awareness through foreground/background decoupling and fusing. The diversity of instances is further enhanced thanks to the adoption of the non-linear transformation. The size of instances augmented by our method would also be more flexible.

segmentation. Saliency object detection provides only the mask area of salient objects in an image. Thus we need to crop the region (R_n) of novel class objects from d_{syn} . The object mask (M_n) of R_n is obtained using an unsupervised saliency detector [32]. We then paste the object saliency mask M_n onto the background mask $M_{I(s)}$ to obtain the novel class mask of the input image I as a prior.

Although the mask $M_{I(s)}$ obtained through saliency detection provides some object information, it is not semantic-level supervision. Therefore, we adopt the contrastive loss [43] as the loss of Mask Head to learn pixel-level representation, such that pixels of the same objects are pulled in the feature space, while the features of different objects are pushed apart. In this way, a pixel embedding space is introduced as a dense semantic representation to get the semantic segmentation mask M_I of the novel class.

Finally, we assemble the segmented novel class instances into the original base class image I_b , as shown in Eqn 2.

$$I_{saliency} = I_{syn} \odot M_I + I_b \odot (1 - M_I) \quad (2)$$

where $\odot$ is the dot product. By splicing the novel class objects into the image of I_b to remove the relevant region, we get the $I_{saliency}$ image of $d_{saliency}$ and the corresponding mask M_I . As shown in Figure 4 (col.2), $d_{saliency}$ achieves instance-level augmentation by fusing instances into different backgrounds. We refer to more details of the Mask head in the supplementary materials.

3.4. Object Reconstruction Enhancement Module

In this section, we introduce a non-linear data augmentation method at the instance level, namely the Object Re-

construction Enhancement Module (OREM), which effectively enhances the diversity of C_{novel} instances during finetuning. We draw inspiration from masked image modeling in self-supervised learning, such as Masked Autoencoder (MAE) [14] leverage a self-supervised masked encoder to generate the reconstructed image of the input mask tokens. Thus, we utilize the non-linear fitting capability of the MAE to further augment instance diversity. To maximize the impact of MAE on the object, we crop each novel class object from I and its corresponding mask M to obtain the cropped k_{th} novel class objects image I^k and mask M^k . The center of the object is the crop center, and the crop size is based on the size of the object. To fit the input size of MAE, the length and width are set to the same. The object patch and mask are scaled to 224×224 resolution. The formula of clipping (f_c) and scaling (f_s) is as follows:

$$I_n^k, M_n^k = f_s(f_c(I^k, M^k)) \quad (3)$$

The diversity of samples has been improved by fusing instances into different backgrounds in SAES. Thus we expect that part of the foreground area will also be enhanced while the background area remains unchanged in OREM, so as to increase the diversity of the instance. Specifically, we divide the I_n^k image into N non-overlapping patches ($\{x_i^p\}_{i=1}^N$) and the positions of patches are denoted as $P \in \{1, \dots, N\}$. All patches are divided into patches of the novel classes foreground and other patches. Their positions are denoted as $P_f \in \{1, \dots, N\}^J$ and $P_b \in \{1, \dots, N\}^K$ respectively, where J and K represent the number of foreground and other patches respectively, and $K + J = N$. Then we randomly mask the proportion of the foreground patch of novel classes. The masked positions are denoted $M_f \in \{1, \dots, N\}^{k\%J}$, where $k\%$ is the masking ratio of foreground patches. The remaining patches x_M^k are shown in Eqn 4.

$$x_M^k = \{x_i^p : i \in P_b\}_{i=1}^N \cup \{x_i^p : i \notin M_f \cap i \in P_f\}_{i=1}^N \quad (4)$$

$$I_{mae}^k = f_D(f_E(x_M^k)) \quad (5)$$

As illustrated in Eqn 5, the remaining patches x_M^k are sent to the pretrained encoder f_E and decoder f_D of MAE to generate the reconstructed image of the novel class I_n^k , which can serve as the enhanced version of the object. Finally, we will paste the enhanced object I_{mae}^k back to $I_{saliency}$ to generate I_{mae} of d_{mae} (Figure 4, col.3).

Figure 5 shows the visualizations of the results of Cutout [5], GridMask [3], CutMix [54], and our method on Pascal VOC. Unlike the above methods, our method obtains semantic awareness through foreground/background decoupling and fusing. The diversity of instances in our method are further enhanced thanks to the adoption of the non-linear transformation. The size of instances augmented by our method would also be more flexible.

Method/Shot		Novel Split 1					Novel Split 2					Novel Split 3					Mean
		1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
FSRW [18]	ICCV 19	14.8	15.5	26.7	33.9	47.2	15.7	15.3	22.7	30.1	40.5	21.3	25.6	28.4	42.8	45.9	28.4
MetaDet [45]	ICCV 19	18.9	20.6	30.2	36.8	49.6	21.8	23.1	27.8	31.7	43.0	20.6	23.9	29.4	43.9	44.1	31.0
Meta R-CNN [51]	ICCV 19	19.9	25.5	35.0	45.7	51.5	10.4	19.4	29.6	34.8	45.4	14.3	18.2	27.5	41.2	48.1	31.1
MPSR [49]	ECCV 20	41.7	-	51.4	55.2	61.8	24.4	-	39.2	39.9	47.8	35.6	-	42.3	48.0	49.7	35.8
CGDP+FSN [27]	CVPR 21	40.7	45.1	46.5	57.4	62.4	27.3	31.4	40.8	42.7	46.3	31.2	36.4	43.7	50.1	55.6	43.8
FSCE [41]	CVPR 21	44.2	43.8	51.4	61.9	63.4	27.3	29.5	43.5	44.2	50.2	37.2	41.9	47.5	54.6	58.5	46.6
SRR-FSD [58]	CVPR 21	47.8	50.5	51.3	55.2	56.8	32.5	35.3	39.1	40.8	43.8	40.1	41.5	44.3	46.9	46.4	44.8
CME [26]	CVPR 21	41.5	47.5	50.4	58.2	60.9	27.2	30.2	41.4	42.5	46.8	34.3	39.6	45.1	48.3	51.5	44.4
TIP [24]	CVPR 21	27.7	36.5	43.3	50.2	59.6	22.7	30.1	33.8	40.9	46.9	21.7	30.6	38.1	44.5	50.9	38.5
QA-FewDet [11]	ICCV 21	42.4	51.9	55.7	62.6	63.4	25.9	37.8	46.6	48.9	51.1	35.2	42.9	47.8	54.8	53.5	48.0
FSOD-UP [48]	ICCV 21	43.8	47.8	50.3	55.4	61.7	31.2	30.5	41.2	42.2	48.3	35.5	39.7	43.9	50.6	53.5	45.0
FADI [1]	NeurIPS 21	50.3	54.8	54.2	59.3	63.2	30.6	35.0	40.3	42.8	48.0	45.7	49.7	49.1	55.0	59.6	49.2
Meta Faster R-CNN [12]	AAAI 22	43.0	54.5	60.6	66.1	65.4	27.7	35.5	46.1	47.8	51.4	40.6	46.4	53.4	59.9	58.6	50.5
KFSOD [56]	CVPR 22	44.6	-	54.4	60.9	65.8	37.8	-	43.1	48.1	50.4	34.8	-	44.1	52.7	53.9	49.2
FCT [13]	CVPR 22	49.9	57.1	57.9	63.2	67.1	27.6	34.5	43.7	59.2	51.2	39.5	54.7	52.3	57.0	58.7	51.6
LVC [20]	CVPR 22	54.5	53.2	58.8	63.2	65.7	32.8	29.2	50.7	49.8	50.6	48.4	52.7	55.0	59.6	59.6	52.3
TENET [55]	ECCV 22	46.7	-	55.4	62.3	66.9	40.3	-	44.7	49.3	52.1	35.5	-	46.0	54.4	54.6	50.7
MRSN [31]	ECCV 22	47.6	48.6	57.8	61.9	62.6	31.2	38.3	46.7	47.1	50.6	35.5	30.9	45.6	54.4	57.4	47.7
TFA w/ cos [46]	ICML 20	39.8	36.1	44.7	55.7	56.0	23.5	26.9	34.1	35.1	39.1	30.8	34.8	42.8	49.5	49.8	39.9
Ours (TFA w/ cos)		42.3	44.8	45.7	56.9	57.2	26.7	31.5	36.8	37.2	40.5	33.0	37.1	43.9	50.5	55.4	42.6
DeFRCN [34]	ICCV 21	53.6	57.5	61.5	64.1	60.8	30.1	38.1	47.0	53.3	47.9	48.4	50.9	52.3	54.9	57.4	51.9
Ours (DeFRCN)		59.0	60.6	63.8	64.6	65.2	34.7	40.6	48.1	54.1	49.9	50.7	52.0	55.4	58.3	61.8	54.6

Table 1. Few-shot detection performance across the 3 splits on the PASCAL VOC benchmark. The best and second-best results are colored red and blue, respectively. Two methods equipped with our idea consistently outperform relevant baselines under all settings and achieve competitive performance compared with recent state-of-the-art methods for 3 novel splits.

4. Experiments

4.1. Few-Shot Object Detection Benchmarks

We followed the previous work [46] to use data splits and the novel samples to evaluate our model on two widely used few-shot detection datasets: PASCAL VOC [6] and MS-COCO [29] for a fair comparison.

PASCAL VOC have three different data splits, where each split group randomly divides 20 classes into 15 base classes and 5 novel classes. For each novel category, there are K=1, 2, 3, 5, and 10 shots available from the combination of VOC2007 and VOC2012 train/val sets for finetuning. We evaluate the performance on the VOC2007 test set with the standard PASCAL VOC metric, Average Precision (IoU=0.5), and report it as nAP50 for the novel categories.

MS-COCO comprises 80 categories, of which 20 are considered novel categories in common with PASCAL VOC, and the remaining 60 belong to the base classes. To finetune the model, we use K=1, 2, 3, 5, 10, and 30 shots for each novel class. We evaluate the performance on 5k images from the validation set. We report the COCO-style AP as the evaluation metric for novel categories.

4.2. Implementation Details

In this paper, our experiments are conducted on two robust baselines: TFA [46] and DeFRCN [34]. TFA is a simple effective finetuning-based method, which adopts a two-stage scheme and only finetunes the last layer of de-

tectors, which is comparable to meta-based methods. DeFRCN is a recent SOTA method, which looks closely into the visual task of FSOD and proposes a simple yet effective finetuning-based framework. We evaluate DeFRCN performance (mAP) over multiple runs.

Traditional data augmentation is employed in CutMix to prevent overfitting, including random scaling, clipping, and flipping, due to novel classes with only a few samples. We adopt SGD optimization, with the momentum and weight attenuation of 0.9 and 0.0001 respectively. The learning rate during base training is set to 0.02, and the learning rate is set to 0.01 during few-shot finetuning. Moreover, the proportion k_1 , k_2 , and k_3 of different generated sets is set to 0.3, 0.3 and 0.4 respectively. The masking ratio of foreground patches is set to 30%. All experiment was carried out on 4 Nvidia GeForce RTX 3090 GPUs with a batch size of 16 by the open-source library PyTorch (<https://pytorch.org/>).

4.3. Comparison Results

Results on PASCAL VOC. Table 1 presents the results from 3 novel splits of PASCAL VOC compared with baselines and the existing state-of-the-art methods. It is obvious that our few-shot data augmentation method surpasses the two baselines in all splits and shots. Our method based on DeFRCN [34] achieves the best performance with 54.6%, which outperforms the baseline with a margin of 2.7 on average. In terms of overall performance, our method demonstrates superior performance compared to most existing

Method		Shot Number						
		1	2	3	5	10	30	
Meta R-CNN [51]	ICCV 19	1.0	1.8	2.8	4.0	6.5	11.1	
MPSR [49]	ECCV 20	5.1	6.7	7.4	8.7	9.8	14.1	
FSDetView [50]	ECCV 20	4.5	6.6	7.2	10.7	12.5	14.7	
FSCE [41]	CVPR 21	-	-	-	-	11.9	16.4	
Retentive R-CNN [8]	CVPR 21	-	-	-	-	10.5	13.8	
SRR-FSD [58]	CVPR 21	-	-	-	-	11.3	14.7	
CME [26]	CVPR 21	-	-	-	-	15.1	16.9	
DCNet [17]	CVPR 21	-	-	-	-	12.8	18.6	
TIP [24]	CVPR 21	-	-	-	-	16.3	18.3	
QA-FewDet [11]	ICCV 21	4.9	7.6	8.4	9.7	11.6	16.5	
FSOD-UP [48]	ICCV 21	-	-	-	-	11.0	15.6	
LVC [20]	CVPR 22	-	-	-	-	17.8	24.5	
KFSOD [56]	CVPR 22	-	-	-	-	18.5	-	
FCT [13]	CVPR 22	5.6	7.9	11.1	14	17.1	21.4	
Meta Faster R-CNN [12]	AAAI 22	5.1	7.6	9.8	10.8	12.7	16.6	
MRSN [31]	ECCV 22	-	-	-	-	15.7	17.5	
TFA [46]	ICML 20	4.4	5.4	6.0	7.7	10.0	13.7	
Ours (TFA)		5.6	6.2	7.5	8.6	11.2	14.3	
DeFRCN [34]	ICCV 21	9.3	12.9	14.8	16.1	18.5	22.6	
Ours (DeFRCN)		10.2	14.5	15.8	16.9	19.1	23.1	

Table 2. Experiments on COCO dataset. The best and second-best results are colored red and blue, respectively. Two methods equipped with our idea consistently outperform relevant baselines under all settings. A new state-of-the-art result is also achieved.

methods across different splits and shots on average, which verifies the robustness and generalization of our method.

Results on MS-COCO. For the MS COCO dataset, we adopt some recent works for comparison, as shown in Table 2. After applying our method, we consistently outperform baselines and achieve 13% and 6% improvement over the two baselines on average respectively. And new state-of-the-art results under different shot settings are achieved with other existing methods. Under the setting of 1, 2, 3, 5, 10, and 30 shots, our method achieves 10%, 12%, 6%, 5%, 4%, and 3% performance gain compared with DeFRCN [34] on AP respectively. Especially under the setting of 1-2 shot instances is quite limited, our method is greatly improved with a boost of up to about 10% AP. The results show that our data augmentation method performs better on the more difficult MS COCO dataset.

4.4. Ablation Studies

To investigate the proposed method’s effectiveness, we conducted three ablation experiments, namely, the “Impact of the Semantic Awareness Enhancement Strategy”, the “Impact of the Object Reconstruction Enhancement Module”, and the “Impact of the Proportion of Different Generated Sets”. DeFRCN [34] is adopted as the baseline for ablation studies on the Novel Split 1 of PASCAL VOC. Table 3 summarizes the results of our method under different settings, and the following is a detailed comparison. Note that we adopt CutMix data augmentation to increase the

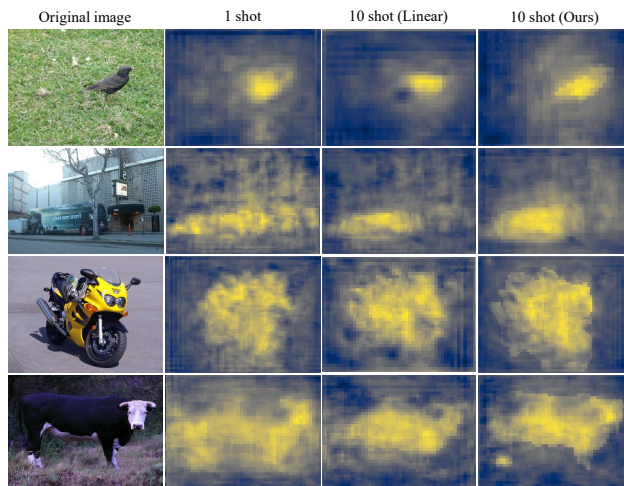


Figure 6. Visualizations of confidence weighting maps with different categories in novel split 1 of PASCAL VOC. “10 shot (Linear)” and “10 shots (Ours)” indicates to enhance instances from 1-shot to 10-shot with linear and proposed methods respectively. Comparing with “10 shot (Linear)”, the distribution of confidence weighting maps of “10 shot (Ours)” better adapts to the shape characteristics of various categories, which means our method could effectively help distinguish foregrounds from backgrounds via diversity enhancement especially only few samples are available.

number of training samples for novel classes and improve the performance by 1.2 on average.

Impact of Semantic Awareness Enhancement Strategy.

In order to assess the impact of the SAES, we incorporated SAES into the baseline model with CutMix and conducted a comparative analysis. The results presented in Table 3 (line 2 vs line 3) demonstrate that SAES has yielded a considerable improvement, indicating that the decoupling of foreground and background is critical for effectively learning objective semantic discriminative features. By fusing instances into different backgrounds, the adoption of SAES can indeed bring performance improvements.

Impact of Object Reconstruction Enhancement Module.

In order to validate the impact of the OREM, we add and compare it to the baseline with CutMix while keeping other conditions unchanged. The experimental results show that the OREM outperforms the baseline with CutMix and improves the overall performance by 1.0 on average (Table 3, line 2 vs line 4). That is, such a non-linear data enhancement procedure is important for further promoting the performance of FSOD since it could better exploit the potential of data by augmenting data diversity.

Moreover, we also carefully analyze the influence of setting different masking ratios $k\%$ in the OREM, which is used to mask random patches of the input novel class instance. As shown in Table 4, different values of $k\%$ are applied in the experiment under novel split 1 of PASCAL

CutMix	SAES	OREM	Novel Split 1					Novel Split 2					Novel Split 3					Mean	Speed (s/iter)
			1	2	3	5	10	1	2	3	5	10	1	2	3	5	10		
			52.8	57.7	59.9	62.4	60.0	30.7	38.7	46.9	52.8	47.9	47.3	50.2	52.5	55.2	58.9	51.6	0.48
✓			55.5	57.6	61.4	63.9	63.5	32.0	38.9	47.4	53.6	48.9	48.8	49.9	53.4	56.7	59.8	52.8	0.50
✓	✓		57.3	59.9	63.7	64.1	63.7	33.7	39.3	47.1	54.4	49.6	49.4	51.3	55.2	58.0	61.0	53.9	0.59
✓		✓	58.1	59.7	63.3	64.3	64.1	32.4	39.3	48.8	54.6	49.5	49.6	50.8	55.1	57.5	60.7	53.8	0.56
✓	✓	✓	59.0	60.6	63.8	64.6	65.2	35.7	40.6	48.1	54.1	49.9	50.7	52.0	55.4	58.3	61.8	54.7	0.62

Table 3. Ablation study of different components of our method for FSOD on 3 novel splits of PASCAL VOC. After finetuning with the SAES, our model achieves a significant improvement over the baseline. The performance could be further promoted once the OREM is adopted. So the effectiveness of our data augmentation is fully played under this setting.

Masking Ratio	Shot					Mean
	1	2	3	5	10	
0%	55.5	57.6	61.4	63.9	63.5	60.4
10%	56.2	59.2	62.6	63.7	64.1	61.1
20%	58.2	60.5	62.9	64.4	65.6	62.3
30%	59.0	60.6	63.8	64.6	65.2	62.6
50%	58.9	59.3	63.6	64.7	64.7	62.2
70%	57.4	57.9	62.9	64.2	64.6	61.4

Table 4. Ablation study of different masking ratios in OREM on novel split 1 of PASCAL VOC. Note the masking ratio here is not for all patches of the entire image, but only for the foreground.

k_1	k_2	k_3	Shot					Mean
			1	2	3	5	10	
1.0	0.0	0.0	55.5	57.6	61.4	63.9	63.5	60.4
0.5	0.5	0.0	58.1	59.7	63.3	64.3	64.1	61.9
0.5	0.2	0.3	58.6	59.3	63.5	64.9	64.8	62.2
0.3	0.3	0.4	59.0	60.6	63.8	64.6	65.2	62.6
0.2	0.2	0.6	58.8	59.4	63.0	64.0	64.5	61.9
0.0	0.0	1.0	58.1	59.4	63.6	63.8	63.8	61.7

Table 5. Ablation study of weights for different sets on novel split 1 of PASCAL VOC. k_1 , k_2 , and k_3 represent the weights of the synthetic set, saliency set, and MAE set respectively.

VOC. It shows that the OREM achieves the best performance at the masking ratio of 30%. Images generated by an undersized masking ratio are lack variety, which affects the generalization of the detection model. However, the excessive masking ratio significantly alters the resulting image, impeding the model convergence. Furthermore, as shown in Figure 6, comparing with traditional linear augmentation methods (10 shot (Linear)), the distribution of confidence weighting maps of our non-linear method (10 shot (Ours)) better adapts to the shape characteristics of different categories, which means our method could effectively help distinguish foregrounds from backgrounds via diversity enhancement especially only a few samples are available.

Impact of the Proportion of Different Generated Sets. We use 6 different weights of k_1 , k_2 , and k_3 to verify the impact of the proportion of different generated sets, as shown

in Table 5. It can be seen that using only one generated set (k_1) could increase the overfitting risk of the model. Thus, it is necessary to balance the weight of different generated sets. When the dataset contains multiple generated sets, the model tends to achieve higher performance (Table 5, line 1 to line 4). It indicates that the model can obtain training opportunities on different generated sets, and the model can better learn all features, thus improving the accuracy. In addition, we find that by adequately increasing the weight of k_2 , k_3 , the model shows better performance when there are only one or two instances. This is due to the enhancement of the diversity of few-shot instances.

4.5. Drawback and Discussion

As shown in the last column of Table 3, we also evaluate the speed of finetuning under all five settings. We can see that despite the significant performance improvement, the proposed method is nearly 29.2% slower than the baseline during the training. But it is worth noting that finetuning-based few-shot object detection models equipped with our idea will not bring any extra computational burden during the inference but consistently achieve better results.

5. Conclusion

In this paper, we propose an instance-level data augmentation method for FSOD, which decouples the foreground and background to increase their semantic diversity respectively. To decouple the foreground background of few-shot instances, the semantic awareness enhancement strategy is adopted to extract the mask of instances and fuse instances into different backgrounds. The object reconstruction enhancement module is additionally introduced to enhance the instances in non-linear transformation spaces to further enhance the instance diversity. The proposed method can be readily plugged into existing finetuning-based FSOD methods for further performance improvement. Extensive experiments on two datasets demonstrate the model equipped with our method significantly outperforms the baseline by a large margin. And new state-of-the-art results on the PASCAL VOC and MS COCO under different shot settings are achieved, demonstrating the effectiveness of our idea.

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